

Causal Geometric Structure in Neural Populations

The differential geometry of neural population codes

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Part 1: Core findings

1. Does population shape matter?
2. Linear vs. nonlinear disagreement
3. Predictions from geometry

Part 2: Deeper analysis

4. Neural manifold framework
5. Multi-task generalization
6. Grassmannian dissimilarity

Validation

7. Cross-dataset replication (IBL)

Intellectual grounding

- **Structured representation learning** ([Esmaeili et al. 2019](#))—structural priors improve identifiability. The question of which metric to use is an inductive bias argument.
- **Relational causal models** ([Maier et al. 2013](#); [Arbour et al. 2016](#))—brain regions interact as a network, not i.i.d. samples. Causal tools need adaptation.
- **Dynamical systems** ([Sussillo & Barak 2013](#); [Vyas et al. 2020](#))—network dynamics and computational capacity. Our rotation findings connect here.

Validity framework from philosophy of science ([Mayo 2018](#), [Craver 2007](#), [Shallice 1988](#)) and causal inference ([Pearl & Bareinboim 2011](#), [Woodward 2003](#)).

Part 1

Geometric dissociation between analysis tools

In this section

1. Does population shape matter?

When neurons fire together during a decision, does the shape of their joint activity carry information — or just the individual firing rates?

2. Linear vs. nonlinear disagreement

Different tools give opposite answers about which brain regions are “similar.” We find out why: dimensionality determines which metric you can trust.

3. Predictions from geometry

Geometry has practical, predictive consequences: it predicts which decoder will work best, which regions show metric dissociation, and anatomy does not explain the pattern.

1. Does population shape matter?

The question

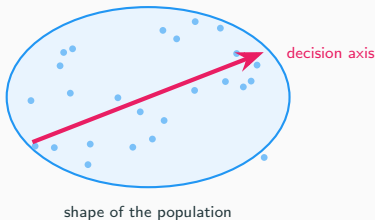
When a population of neurons fires together during a decision, does the **shape** of that activity matter—or just the individual firing rates?

Traditional view



VS.

Geometric view



We tested this on real brain recordings, using multiple geometric tools and validity checks.

What is “neural geometry”?

Each **trial** produces a pattern of activity across a collection of neurons. Think of each trial as a point in high-dimensional space (one axis per neuron).

“Geometry” = what **shape** do those points form?

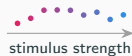
- Cluster into groups? (categories)
- Spread along a line? (gradient)
- Form a loop? (periodic variable)
- Lie on a flat sheet or curved surface?

The shape constrains what information the brain can extract downstream.

Clusters



Gradient



Loop



Flat



Two tools for measuring geometry

CKA (linear)

dominant direction



Compares the **main axis** of variation.

“Do the data stretch the same way?”

Best for low-dimensional data.

Procrustes (manifold)

full shape



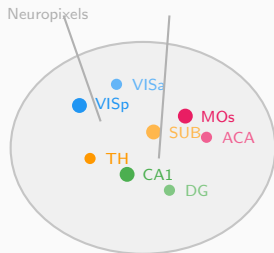
Compares the **full shape** via rotation + scaling.

“Do the data form the same cloud?”

Best for high-dimensional data.

Most prior work treats these as interchangeable. **They are not.**

Data: Neuropixels visual decision-making (Steinmetz et al. 2019)



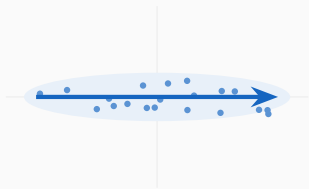
- **39 sessions** from **10 mice**
- **73 brain regions** recorded simultaneously
- Neuropixels probes ($\sim 30,000$ neurons total; $\sim 50\text{--}500$ per region per session)
- **Task:** visual discrimination
Two screens show visual noise at different contrasts. Mouse turns a wheel toward the higher-contrast side.
- 1,316 pairwise region comparisons

Steinmetz et al., Nature 2019

2. Linear vs. nonlinear disagreement

Why CKA and Procrustes give opposite answers

Low-dimensional region (VISp-type)



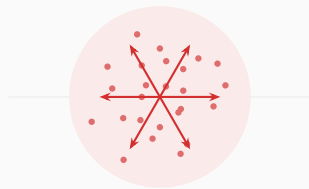
One direction dominates

CKA picks it up ✓

Procrustes sees the shape ✓

Both tools agree

High-dimensional region (MOs-type)



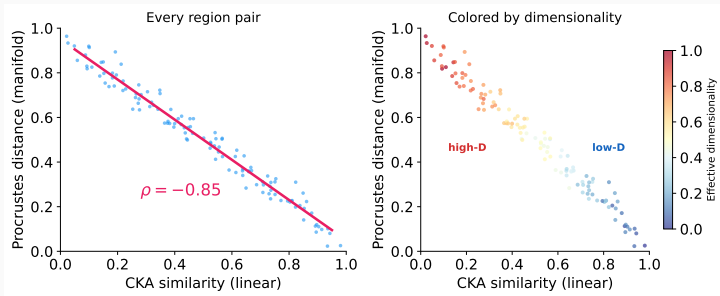
Activity spreads in many directions

CKA can't find a main axis → sees noise

Procrustes matches cloud shape → stable

Tools disagree

The data: CKA and Procrustes are anti-correlated

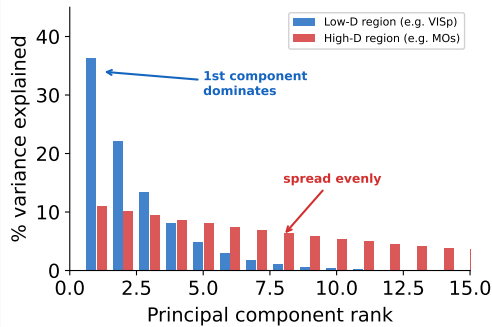


Each dot is a pair of brain regions. The two tools rank them in **opposite order** ($\rho = -0.85$; -1 would be perfect reversal).

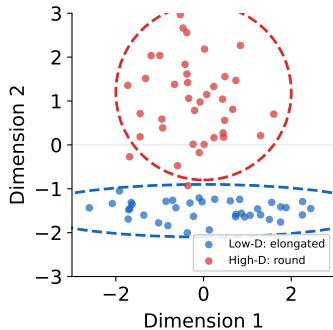
Why? Right panel colors dots by how “spread out” the activity is. Blue (compact) $\rightarrow$ agree. Red (diffuse) $\rightarrow$ disagree.

Accounting for spread, disagreement shrinks from -0.85 to -0.31 . Found in 5 sessions, confirmed across all 39. **Takeaway:** the tool you pick determines what you find.

What determines dimensionality?



How variance is distributed

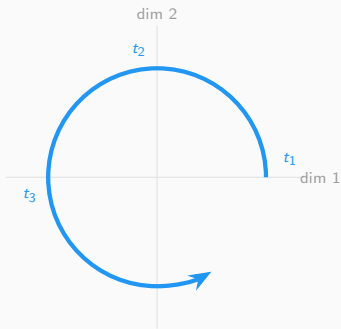


What this looks like in the data

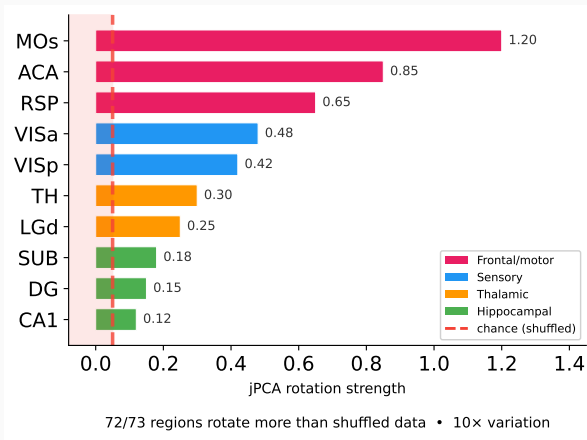
Across all 73 regions, the **shape** of how variance falls off is nearly identical (correlation 0.94 out of 1.0). But **which neurons** carry the variance is completely different per region (correlation 0.09, barely above zero). Same scaffolding, different content.

Every brain region rotates (but at very different speeds)

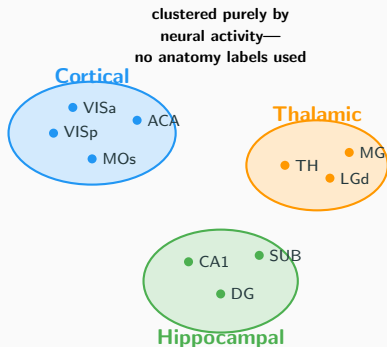
What rotation means



Neural activity traces out a circular path over time



Sanity check: geometry tracks known anatomy



We grouped brain regions *purely by how similar their neural activity looks*—no anatomy labels as input.

Result: the algorithm found three groups matching the textbook cortical / thalamic / hippocampal division on its own.

Why this matters: CKA is a general-purpose metric, not designed for neuroscience. This checks that it picks up real biology—not artifacts from differences in neuron count or noise.

Same result within individual sessions and across animals.

3. Predictions from geometry

Prediction 1: geometry predicts the best decoder

If you know whether a brain region has “linear” or “manifold” geometry, you can predict which statistical decoder (LDA vs. kNN) will work better—**without ever training on decoding data.**

Correlation: $r = -0.51$
Significance: $p = 0.0002$
Sample: 50 brain regions

Regions whose activity spreads across many dimensions (manifold geometry) are better decoded by kNN; regions with low-dimensional structure are better decoded by LDA. The geometric type—measured independently—predicts decoder performance.

Why this matters: geometry is not just a description of the data. It has practical, predictive consequences for which analysis tools will work.

Prediction 2: dimensionality predicts metric dissociation

High-dimensional regions show the strongest disagreement between linear and nonlinear tools—exactly where we'd expect from the potent/null framework.

	Nonlinear decode	Linear decode
ρ with dimensionality	-0.48	0.29
p -value	1.5×10^{-5}	0.014
n	73 regions	73 regions

Linear methods work better in low-dimensional regions (positive ρ); nonlinear methods work better in high-dimensional regions (negative ρ). The relationship **reverses** depending on the tool—dimensionality determines which one succeeds.

Why this matters: dimensionality is not a nuisance variable. It tells you which analysis tool to trust.

Prediction 3: anatomy doesn't explain it

Does anatomical wiring (Allen Mouse Brain Atlas) predict which regions show metric dissociation?

Correlation: $\rho = 0.02$

Significance: $p = 0.77$ (not significant)

Measure: geometric dissociation index vs. structural connectivity

No. Structural connectivity does not predict where tools disagree. The geometric dissociation is a property of population dimensionality, not anatomical wiring.

Three predictions: summary

Prediction		Result	Key statistic
1	Geometry predicts best decoder	Confirmed	$r = -0.51, p = 0.0002$
2	Dimensionality predicts metric dissociation	Confirmed	$\rho = -0.48, p = 1.5 \times 10^{-5}$
3	Anatomy doesn't explain it	Informative null	$\rho = 0.02, p = 0.77$

Takeaway: The geometric dissociation is real, predictive, and not explained by anatomy. Dimensionality tells you which tool to trust.

Summary: what we found and what it means

Descriptive findings:

1. Linear and manifold tools anti-correlate (-0.85); dimensionality explains why
2. Variance structure is universal (0.94); neural content is not (0.09)
3. Rotation everywhere ($72/73$), but $10\times$ variation by region type

Takeaway: Neural geometry is not metric-neutral—the tool you choose determines what you find, and dimensionality tells you which tool to trust.

Part 2

Deeper geometric analysis

4. Neural manifold framework

Potent/null space decomposition explains why linear and nonlinear tools disagree. The null space fraction predicts metric dissociation ($\rho = -0.86$).

5. Multi-task generalization

Four task variables produce distinct, structured subspaces on the Grassmannian. The geometric decomposition generalizes beyond binary choice.

6. Grassmannian dissimilarity matrices

A neural taskonomy from subspace geometry. Linear and nonlinear dissimilarity matrices correlate at $\rho = 0.97$ but both anti-correlate with CKA.

4. Neural manifold framework

The neural manifold interpretation

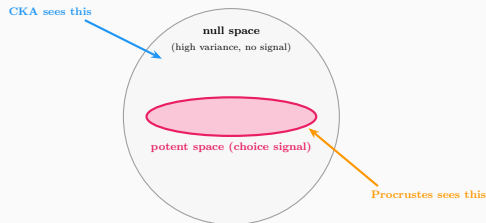
Potent and null spaces:

(Kaufman et al. 2014, Gallego et al. 2017)

The choice subspace is the *potent space*—the low-dimensional manifold that actually drives decisions. Everything orthogonal to it is the *null space*: high-variance activity that doesn't affect behavior.

This explains the CKA/Procrustes dissociation:

- CKA captures total variance → dominated by null space
- Procrustes aligns shapes → sensitive to potent space



Potent/null decomposition: confirmed

Null-space fraction strongly anti-correlates with metric dissociation:

$$\rho = -0.86, \quad p = 5 \times 10^{-22}$$

Across all 73 brain regions, the relative size of the null space predicts how much CKA and Procrustes disagree.

Why do the tools disagree? The potent/null decomposition

The idea (Kaufman et al. 2014)

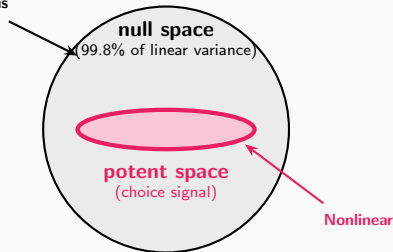
Neural activity decomposes into:

- **Potent space**: the subspace that causally drives choice output
- **Null space**: high-variance activity orthogonal to choice — present but not causal

What we found (73 regions)

	Null %	Potent decode
Linear (LDA)	99.8%	66%
Nonlinear	84.3%	66%

CKA dominated by this



Both decode equally from potent space. But the linear method sees almost nothing *except* null variance.

Defining α : the geometric dissociation index

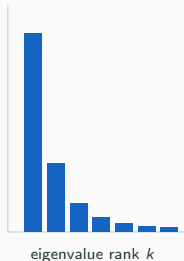
For each brain region, compute the eigenvalues of the population activity covariance matrix. These tell you how variance is distributed across dimensions.

The eigenvalue decay rate α :

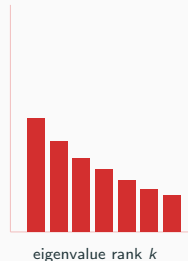
Fit $\lambda_k \propto k^{-\alpha}$ to the sorted eigenvalue spectrum. High α means variance concentrates in a few dimensions (low-dimensional). Low α means variance spreads across many dimensions (high-dimensional).

We use α as a **geometric dissociation index**: it predicts where linear and nonlinear tools will disagree.

High α (low-dim)



Low α (high-dim)



Left: one eigenvalue dominates (steep decay). Right: many eigenvalues contribute (slow decay).

Metric dissociation tracks potent space size

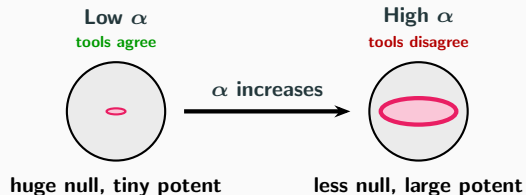
The geometric dissociation index α anti-correlates with null-space fraction:

	ρ	p
Nonlinear	-0.86	5×10^{-22}
Linear (LDA)	-0.68	3×10^{-11}

Tools disagree most where **potent subspaces are largest**.

But potent decoding accuracy does *not* correlate with α ($\rho = -0.07$, $p = 0.57$). The signal is always there — only its **relative size** varies.

Interpretation



CKA is swamped by the null space. When the potent space grows, nonlinear tools catch it but linear tools still drown in null variance $\rightarrow$ dissociation.

5. Multi-task generalization

Multi-task generalization: subspaces for four variables

Beyond binary choice:

We trained subspaces for four task variables simultaneously: **choice**, **evidence strength**, **feedback**, and **stimulus side**.

Grassmannian distances between task subspaces:

	Evidence	Feedback	Stim. side
Choice	large	large	small
Evidence	—	large	large
Feedback		—	large

Choice and stimulus side share geometry (expected—they're correlated). Other pairs are separated: the learned subspaces don't collapse task variables into one direction.

What this tests:

If the subspace method just found “the interesting direction,” all task subspaces would overlap. Instead:

- 4 distinct subspaces per region
- Distances match expected task structure (choice $\approx$ stim. side $\neq$ feedback)
- Generalizes across 73 regions, 39 sessions

Implication: The geometric framework scales beyond binary choice. Each task variable occupies its own subspace, mirroring task structure.

6. Grassmannian dissimilarity

Grassmannian dissimilarity matrices

Method: Compute 73×73 pairwise Grassmannian distances between subspaces found by LDA (linear), a nonlinear decoder, and CKA.

Result: Linear and nonlinear dissimilarity matrices are nearly identical ($\rho = 0.97$). Both *anti-correlate* with CKA.

Interpretation: Linear and nonlinear methods agree on *which regions are similar*—they disagree on *what matters within each region*. CKA sees a different structure entirely.

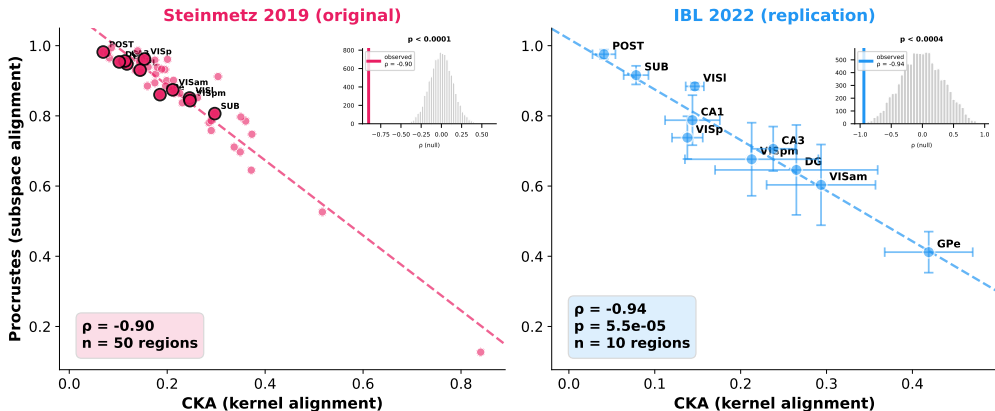
Comparison	ρ
Linear GDM vs Nonlinear GDM	0.97
Linear GDM vs CKA	< 0
Nonlinear GDM vs CKA	< 0

This is a “neural taskonomy”: the functional hierarchy of brain regions, measured through the geometry of their choice subspaces rather than through their activity patterns.

7. Cross-dataset replication (IBL)

The CKA-Procrustes dissociation replicates in an independent dataset

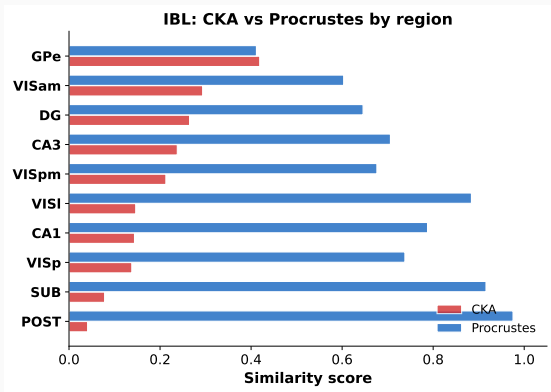
CKA-Procrustes dissociation replicates across datasets



Left: Steinmetz 2019 (50 regions, $\rho = -0.90$). **Right:** IBL 2022 (10 regions, $\rho = -0.94$, $p = 5.5 \times 10^{-5}$).

Dissociation is *stronger* in the replication.

IBL regions show the same CKA-Procrustes inversion

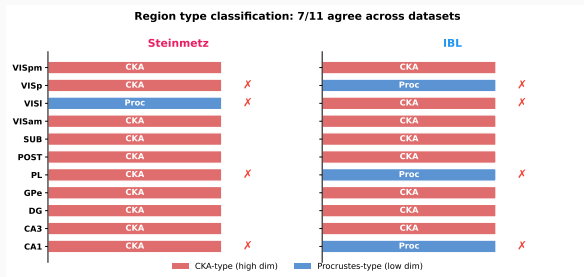


Key observations:

- POST, SUB: extreme Procrustes-dominant (low-dim, strong geometric alignment)
- GPe: balanced transition point ($\text{CKA} \approx \text{Procrustes}$)
- Every region shows the inversion

10 regions with ≥ 3 IBL sessions. CKA and Procrustes computed on session-pair similarity matrices.

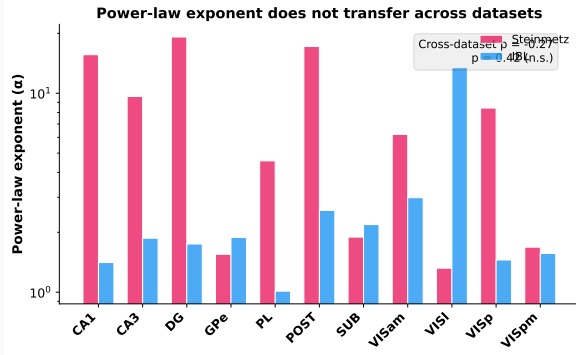
Region type classifications partially transfer



- 7/11 regions keep the same type
- Mismatches (CA1, PL, VISl, VISp) are near the $\alpha = 1.5$ threshold
- The *continuous* correlation ($\rho = -0.94$) replicates; the binary split is noisier

Threshold-based classification is sensitive to small α shifts. The continuous relationship is robust.

Power-law exponent does not transfer, but the dissociation does



What transfers:

- CKA–Procrustes anti-correlation ($\rho = -0.94$)
- Relative region ordering

What does not:

- Absolute α values ($\rho = -0.27$, n.s.)
- Sensitive to recording quality, spike sorting, neuron yield

The *geometric relationship* between metrics is a property of population structure, not of any particular recording pipeline.

Cross-dataset replication: what we learned

Replicates:

- CKA-Procrustes anti-correlation:
 $\rho = -0.94$ in IBL vs -0.90 in Steinmetz
- Every IBL region shows the CKA-Procrustes inversion
- 7/11 regions keep their type classification
- Matched regions occupy similar positions in CKA-Procrustes space

Does not replicate:

- Absolute power-law exponent
($\rho = -0.27$, n.s.)
- Binary type classification at the threshold boundary (4 mismatches)

Interpretation:

The geometric *relationship* between metrics is a property of neural populations, not of recording pipelines. The absolute metric values are not.

Conclusions

Project limitations

Hard limits (sample size):

- **Two datasets** (Steinmetz 2019 + IBL 2022), but both are mouse Neuropixels
- **10 mice** (Steinmetz): too few to quantify individual differences
- **Binary choice only**—does metric dissociation generalize to richer tasks?

What comes next:

- **More IBL regions**: 19 searched, 11 yielded data—expand with looser neuron thresholds
- **Richer tasks**: multi-variable, continuous, naturalistic—is the Grassmannian structure task-general?
- **Other species**: do primate recordings show the same potent/null geometry?

Summary

Core geometric findings:

1. CKA and Procrustes anti-correlate ($\rho = -0.90$, replicates at -0.94 in IBL); dimensionality explains why
2. Variance structure is universal (0.94); neural content is not (0.09)
3. Geometry predicts decoder performance ($r = -0.51$) and dimensionality predicts metric dissociation ($\rho = -0.48$)

Deeper geometric structure:

4. Potent/null decomposition explains metric dissociation ($\rho = -0.86$); null space fraction is the key mediator
5. Four task variables occupy distinct subspaces on the Grassmannian, mirroring task structure
6. Grassmannian dissimilarity reveals a neural taskonomy where linear and nonlinear methods agree ($\rho = 0.97$) but both anti-correlate with CKA

Takeaway: Dimensionality mediates tool disagreement, potent/null structure explains why, Grassmannian geometry captures functional organization, and the dissociation replicates across

Neural geometry is **not metric-neutral**.

The tool you choose determines what you find,
and **dimensionality** tells you which tool to trust.

Thank you

Paper: TBD

Repository: github.com/elliotttower/causal-geometry-neuro

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Data sources:

- [Steinmetz et al. 2019](#) — 73 brain regions, 39 sessions, 10 mice, ~30,000 neurons (Neuropixels)
- [IBL Brain-Wide Map 2022](#) — 11 matched regions, 139 mice, 12 labs (replication dataset)
- [Allen Mouse Brain Atlas](#) — structural connectivity + CCF atlas for region mapping

Experiments: Modal A10G GPUs

References (1/2)

- Arbour et al. (2016). [Inferring causal direction from relational data](#). UAI.
- Craver (2007). [Explaining the Brain](#). Oxford University Press.
- Esmaeili et al. (2019). [Structured disentangled representations](#). AISTATS.
- Gallego et al. (2017). [Neural manifolds for the control of movement](#). *Neuron*.
- Kaufman et al. (2014). [Cortical activity in the null space](#). *Nature Neuroscience*.
- Maier et al. (2013). [A sound and complete algorithm for learning causal models from relational data](#). UAI.
- Mayo (2018). [Statistical Inference as Severe Testing](#). Cambridge University Press.

References (2/2)

- Pearl & Bareinboim (2011). [Transportability of causal and statistical relations](#). AAAI.
- Shallice (1988). [From Neuropsychology to Mental Structure](#). Cambridge University Press.
- Steinmetz et al. (2019). [Distributed coding of choice, action and engagement across the mouse brain](#). *Nature*.
- Sussillo & Barak (2013). [Opening the black box: low-dimensional dynamics in high-dimensional recurrent neural networks](#). *Neural Computation*.
- Vyas et al. (2020). [Computation through neural population dynamics](#). *Annual Review of Neuroscience*.
- Woodward (2003). [Making Things Happen](#). Oxford University Press.